

signals4.1.sce X signals 4.2.sce X signals 5.1.sce X signals 5.2.sce X signals 6.1.sce X 6.2.sce X signals 7.1.sci X 7.2.sce X 8.1.sce X 8.2.sci X 9.1.sce

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1 // Test Case: Analysis of a Discrete-Time Linear System
2 // Define parameters
3 N = 20; n = 0:N-1; x_n = ones(1, N); y_n = zeros(1, N); // Number of samples
4 // Discrete time index vector
5 // Input sequence: Unit step sequence
6 // Output sequence initialization
7 // Define the difference equation:  $y[n] = 0.5y[n-1] + x[n]$ 
8 for k = 2:N
9     y_n(k) = 0.5 * y_n(k-1) + x_n(k);
10 end
11 // Plot the input sequence
12 clf; // Clear the current figure
13 subplot(2,1,1);
14 plot2d(n, x_n);
15 xlabel('n (Discrete-Time Index)');
16 ylabel('Amplitude');
17 title('Input Sequence: Unit Step Sequence');
18 // Plot the output sequence
19 subplot(2,1,2);
20 plot2d(n, y_n);
21 xlabel('n (Discrete-Time Index)');
22 ylabel('Amplitude');
23 title('Output Sequence: Response of the Discrete-Time System');
24
```



